

Serious Breaches in Clinical Trials

Applicable GxP Discipline(s)

GCP, Regulatory Affairs

What happened?

In a pediatric study a programming error (using the Interactive Response Technology, IRT) occurred. As a result a number of subjects who were already in the active treatment arm were not switched to placebo treatment as per protocol, but continued receiving active treatment.

Specifically, 2 subject groups were receiving placebo and active treatment, respectively. Per protocol, a switch of medication was supposed to occur at week 13 whereby non-responder placebo groups should have received active treatment. To maintain the blind, subjects on active treatment should have received placebo doses at visit weeks 13, 14 and 15 and then continued on active treatment starting at week 16. The IRT error resulted in the active treatment group subjects who reached the week 13 visit receiving active treatment instead of placebo.

The incident affected 18 countries - Argentina, Belgium, Brazil, Colombia, Egypt, Estonia, France, Germany, Guatemala, Hungary, Italy, Latvia, Poland, Spain, Romania, UK, Switzerland and Japan. It was investigated and resolved in a timely manner, but Serious Breach reporting was not performed.

Was this the right thing to do? Should this have been reported as a Serious Breach? Find out more!

Regulatory Background

What is a Serious Breach?

A Serious Breach is a deviation from a clinical trial protocol, Good Clinical Practice (GCP) or any regulatory requirement, which is **LIKELY** to affect to a significant degree:

- a) The safety, rights and physical or mental integrity of any subjects in a clinical trial; or
- b) The reliability and robustness of the data generated in the clinical trial.

It is the responsibility of the sponsor to assess the potential impact of the breach on the safety of trial subjects and the scientific value of the clinical trial to enable reporting to health authorities, where required. The scientific value of the clinical trial depends on a variety of factors such as the study design, type and extent of the data affected by the breach, the overall contribution of the data to key analysis parameters or the impact of excluding the data from analysis.

(For more information on the definition of a Serious Breach please refer to the links to available regulations and guidelines in "Further References").

When is a deviation considered a Serious Breach?

Deviations occur commonly in clinical trials, even with the best designed clinical protocols and investigator training. The majority of these non-compliance instances

are technical deviations that do not result in harm to the trial subjects or significantly affect the scientific value of the reported results of the trial.

These cases should be recorded e.g. in the case report form for the trial or trial master file (TMF), in order for appropriate corrective and preventative actions to be taken. In addition, these protocol deviations should be assessed for inclusion when the clinical study report is produced, as they may have an impact on the analysis of the data. Refer to SOP-7035752 for more information on protocol deviations.

In cases where a deviation has **the potential to significantly impact a trial subject's safety, rights or physical/mental integrity or the reliability and robustness of the data generated in the trial**, it has to be reported as a Serious Breach in some countries, like UK, Spain, Singapore, France, Denmark, Sweden or Australia¹.

In countries that do not have Serious Breach reporting regulations, there may still be a responsibility in identifying and escalating potential Serious Breaches. If a non-Serious Breach regulated country is participating in a global clinical trial and a Serious Breach occurs, that Serious Breach will need to be reported to countries with reporting regulations if that country is also participating in the same clinical trial. The Serious Breach may need to be reported **regardless of where it occurred**, in order to allow HA reporting requirements to be fulfilled in those countries.

1) A complete list of countries and their local requirements is currently compiled by Regulatory Affairs.

Examples

- Examples of Serious Breaches may include dosing errors, inclusion of ineligible subjects into a trial, administration of expired IMP – if subject safety and/or the reliability or robustness of trial data are affected.
- For example, the UK MHRA requires all confirmed instances of clinical fraud to be considered as a Serious Breach. For this type of case national legislation must be taken into consideration with reference to criminal acts such as fraud.
- Failure to appropriately manage safety events may also constitute a Serious Breach if, for example, delays in safety reporting lead to the safety of other patients in the trial being put at risk. In this case, a Serious Breach notification has to be initiated in addition to the usual SAE or SUSAR reporting to health authority.
- For more examples please have a look into the (not exhaustive) list of Serious Breaches in the “Annex” to this article.

When to report?

In countries with a reporting obligation, a Serious Breach should be reported to the health authority without undue delay and no later than **7 calendar days** from becoming aware of the potential breach.

In the countries where reporting is required, a Serious Breach has to be reported as soon as the sponsor or anyone that has a contractual agreement with the sponsor (e.g. CROs, contractors, co-development partners) receives information that provides reasonable grounds to believe that the deviation or non-compliance meets the definition of a potential Serious Breach. If a principal investigator is aware that a Serious Breach has occurred, this information should be promptly reported to the sponsor so that the sponsor can fulfil the reporting requirements.

The sponsor is not expected to wait until all information is available and/or the investigation of the issue is complete; updates can be made as further information becomes available.

Applicable regulations and guidelines

- Serious Breach reporting has been implemented in various countries²⁾ (e.g. in UK since 2006, France and Spain since 2016, Singapore since 2015, Denmark since 2013 and Sweden since 2011).
- In Europe, the requirement for Serious Breach reporting will be implemented across all EU/EEA countries when the Regulation (EU) No 536/2014 (the “Clinical Trial Regulation”) becomes effective in 2019 (planned).
- In the US, sponsors are responsible for reporting and taking actions following the identification of “*unanticipated problems involving risk to human subjects or others*”.
- In addition, ICH E6, R2 addendum states:
 - 5.20.1 addendum - “If noncompliance that significantly affects or has the potential to significantly affect human subject protection or reliability of trial results is discovered, the sponsor should perform a root cause analysis and implement appropriate corrective and preventive actions”
 - 5.20.2 - “If the monitoring and/or auditing identifies serious and/or persistent noncompliance on the part of an investigator/institution, the sponsor should terminate the investigator's/ institution's participation in the trial. When an investigator's/institution's participation is terminated because of noncompliance, the sponsor should notify promptly the regulatory authority(ies).”

2) A complete list of countries and their local requirements is currently compiled by Regulatory Affairs.

Applicable Novartis SOPs

- SOP-7012720 “Management of Scientific Misconduct and Serious GCP Non-Compliance”
- SOP-7038922 “Novartis GOP: Quality/GxP Escalation and Incident Management”

A new GOP is currently under development to further detail management of Serious Breach and ensure compliance with current regulations (see also “What's Next?”).

Back to our example

The investigation performed by Novartis concluded that there was no safety impact to the subjects due to the unintentional overdose, therefore the issue was initially not considered as reportable as a Serious Breach. However, because of the potential harm to the pediatric patients and because of potential impact on the scientific value of the trial data, this error/deviation meets the criteria of a Serious Breach and should have been reported to the applicable health authorities, which finally has been done by Novartis, albeit late.

What's next?

- An e-learning about the general requirements for Serious Breach will be rolled out to all individuals involved in clinical trials at the beginning of next year. Please look out for this!
- The e-learning campaign will be followed by the release of a GOP on Serious Breach including relevant training in the second quarter of 2018.

What do I need to do?³⁾

- Look out for Serious Breaches in your trial, regardless of whether your country has a Serious Breach regulation in place or not.
 - Be aware that some countries require information on Serious Breaches occurring in trial subjects in other participating countries, if the Serious Breach is likely to have a significant impact on the safety or integrity of trial subjects in their own country or if the breach affects the overall scientific value of the trial.

- For example, if a global clinical trial is being conducted in multiple countries, 3 of which happen to be Brazil (no requirement), Spain and the UK (both Serious Breach countries), and a potential Serious Breach occurs in Brazil, both Spain and the UK need to be made aware of the Serious Breach to assess if it meets criteria for reporting to their local health authorities.
- If you discover a deviation which meets the criteria for a Serious Breach, notify the global trial leader **and** your functional Quality partner of the possibility of a Serious Breach as soon as possible:
 - This includes Novartis Country Quality, Clinical QA in GDQ, NIBR Quality and/or NBS/PLS Quality. If you are unsure whom to notify, contact Compliance & RA Quality in GDQ (Stacey Pipala and/or Rika Kremmer).
- Determine with your Quality partner, the global clinical team and other key stakeholders (e.g. RA, Safety, Medical, etc) if the deviation meets the criteria of a Serious Breach:
 - Assess, in cooperation with the principal investigator, whether the deviation is **likely** to have a **significant impact on the safety, rights, physical or mental integrity of subjects**.
 - Assess the impact of the deviation **on the scientific value** of the clinical trial.
 - Document the assessment of whether or not the deviation constitutes a Serious Breach, and archive in the TMF. The documentation and the appropriateness of the decisions taken by the sponsor may be examined during a GCP inspection.
- If a potential Serious Breach is confirmed, notify the country organizations which have Serious Breach reporting obligations and are participating in the trial in a timely manner. Country organizations are to report the breach as per their local procedures and requirements.
- Ensure that IRB/ECs and Chief/Principal Investigators are also notified per local regulations and procedures.
- Escalate the Serious Breach per SOP-7038922 and create a corresponding deviation in AQWA. Complete an investigation to determine the root cause of the Serious Breach and develop a CAPA plan.

3) Please note that the description in this section contains recommendations on how to handle a Serious Breach based on applicable regulations and guidelines. This does not represent the text in current Novartis procedures, but procedures are in the process of being updated.

Takeaway message

It is crucial to identify any Serious Breach on a global basis, even in trials running in countries without a Serious Breach requirement, because other countries may have reporting requirements for Serious Breaches occurring abroad. By assessing and handling them promptly Novartis helps to facilitate the safety of trial subjects, ensures transparency with clinical trial data integrity issues and maintains compliance with applicable law.

More information needed?

For further information or questions, please contact [Rika Kremmer](#), GDQ Knowledge Management Lead. We will forward your questions to the appropriate subject matter expert.

Annex

- Further references & links
- List of examples

Serious Breaches in Clinical Trials - ANNEX

Further references & links

- **MHRA (UK)** - Regulation 29A of the Medicines for Human Use (Clinical Trials) Regulations 2004 [Statutory Instrument 2004/1031], as amended by Statutory Instrument 2006/1928:
<http://www.legislation.gov.uk/ukSI/2006/1928/regulation/16/made>
- **MHRA (UK)** - Guidance for the notification of Serious Breaches of GCP or the trial protocol
https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/404588/GCP_serious_breaches_guide.pdf
- **HSA (Singapore)** - Clinical trials guidance notification of Serious Breaches (HSA):
http://www.hsa.gov.sg/content/dam/HSA/HPRG/Clinical_Trials/Guidance/HSA_CTB_Guidance_Serious_Breach_2May2017.pdf
- **AEMPS (Spain)** - Royal Decree 1090/2015 (English translation): https://www.aemps.gob.es/legislacion/espana/investigacionClinica/docs/Royal-Decree-1090-2015_4-December.pdf
- **ICH** - ICH E6 (R2) Good Clinical Practice (GCP) Guidelines:
http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Efficacy/E6/E6_R2__Step_4.pdf
- **EMA (EU/EEA)** – NOT EFFECTIVE: Guideline for the notification of Serious Breaches of Regulation (EU) No 536/2014 or the clinical trial protocol: http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2017/05/WC500228199.pdf

List of Examples

All examples are taken from the MHRA and HRA homepages and the EMA guidance to Regulation (EU) No 536/2014.

1) The Investigational Medical Product (IMP) was administered to the subject via an incorrect route (e.g. IMP was administered via the intravenous route instead of intramuscular route).	Yes , there was significant potential to impact the safety or physical or mental integrity of trial subjects.
2) A subject was dosed with IMP from the incorrect treatment arm. In addition, some months later, the subjects in an entire cohort were incorrectly dosed with IMP three times daily when they should have been dosed once daily.	Yes , • there was impact on the safety or physical or mental integrity of trial subjects or on the scientific value of the trial • this issue was systematic and persistent leading to a constant breach of the conditions and principles of GCP in connection with that trial or the trial protocol • this issue persisted despite the implementation of a corrective and preventative action plan.
3) One subject was administered 6 additional doses of IMP. The subject was to receive IMP on day 1 and 8 but instead received IMP on days 1 to 8. The subject experienced a severe adverse event as a result.	Yes , there was impact on the safety or physical or mental integrity of trial subjects and on the scientific value of the trial
4) A subject took IMP that had expired two days ago. The subject did not experience any adverse events and this issue was not likely to affect the data credibility of the trial.	No , there was no impact on the safety or physical or mental integrity of the trial subject or on the scientific value of the trial. In addition, the assessment of the breach identified this as a single episode and a detailed corrective and preventative action plan was implemented. Remember that this should be reported as a deviation to the protocol.
5) One subject was administered additional doses of IMP. The subject was given instructions to take higher doses of IMP than what was stipulated in the protocol. The subject experienced a severe adverse event as a result.	Yes , there was impact on the safety of trial subjects and on the scientific value of the trial.
6) Should IMP temperature excursions be reported?	Yes , if the situation was not managed and subjects were dosed with IMP assessed as unstable, which resulted in harm/potential to harm subjects. No , if the excursions had been managed appropriately (e.g. IMP was moved to alternative location/quarantined as necessary and an assessment (by qualified personnel) illustrated that there was no impact on subject safety and data integrity.
7) Multiple issues with the Interactive Response Technology (IRT) system across several clinical trials leading to the dispensing of expired IMP and a shortage of IMP at investigator sites in time of subject visits.	Yes , there was impact on the safety or physical or mental integrity of trial subjects and this issue persisted leading to a constant breach of the conditions and principles of GCP in connection with that trial or the trial protocol, despite the implementation of a corrective and preventative action plan.
8) Due to an Interactive Response Technology (IRT) malfunction 50% of subjects assigned to one arm were unblinded in a blinded trial, furthermore this information was submitted to all trial staff at all investigator sites participating in the trial.	Yes , this could potentially affect the safety of trial subjects, and this was a systematic issue. Yes , this impacts the robustness and reliability of the data generated.

9) On two separate occasions the Sponsors identified issues with the same organization. First with consenting and then with potential fraud in recruitment and consenting. However, there was not unequivocal evidence of fraud at the time of reporting. One of the studies involved pediatric subjects.	Yes , this subsequently led to enforcement action against the organization in question. All confirmed instances of clinical fraud are considered to be a Serious Breach.
10) Concerns were raised during monitoring visits about changes to source data for a number of subjects in a trial, which subsequently made subjects eligible with no explanation. An audit was carried out by the Sponsor and other changes to source data were noted without explanation, potentially impacting on data integrity. Follow-up reports sent to HA confirmed the Sponsor concerns over consenting and data changes made to source without an adequate written explanation.	Yes <i>Note: not all of the information was provided in the original notification, the Sponsor provided follow-up updates.</i>
11) A clinical trial subject attended the hospital emergency department, that attempted to contact the hospital (using the phone number listed on the emergency card issued to the subject) in order to break the unblinding code. Pharmacy was unable to code break in a timely manner, as a result, the subject withdrew from the clinical trial feeling unhappy that the pharmacy was not available in an emergency situation.	Yes , as this had significant potential to harm the subject if unblinding would have affected the course of treatment.
12) A cohort had invalid blood samples as they were processed incorrectly. As a result one of the secondary endpoints could not be met. Therefore, a substantial amendment was required to recruit more subjects to meet the endpoint. Subjects were dosed unnecessarily as a result of this error.	Yes , subjects were dosed unnecessarily as a result of this error.
13) Subject safety was compromised because repeat ECGs were not performed, as required by the protocol. Also, there was inadequate QC of the interim safety reports used for dose escalation which has potential for stopping criteria to be missed.	Yes
14) The Investigator failed to report a single SAE as defined in the protocol (re-training provided).	No , if this did not result in other trial subjects being put at risk, and if it was not a systematic or persistent problem. In some circumstances, failure to report a SUSAR could have a significant impact on trial subjects. Sufficient information and context should be provided for the impact to be assessed adequately.
15) The investigator was not clear on the reporting requirements for the trial and was incorrectly classifying events as expected, as they were common events seen with that particular disease.	Yes , incorrect classification of seriousness criteria, therefore SAEs incorrectly classified as AEs or under-reporting of large numbers of SUSARs.
16) The investigator was not documenting all the AEs associated with the trial.	Yes and No , depending on the type of trial, for example inadequate safety reporting in dose escalation studies may impact the decision to escalate to the next dose level and this should be reported

17) Investigator site failed to reduce or stop trial medication, in response to certain laboratory parameters, as required by the protocol. This occurred with several subjects over a one year period, despite identification by the monitor of the first two occasions. Subjects were exposed to an increased risk of thrombosis.	Yes
18) Minor visit date deviation. A common deviation in clinical trials.	No , a minor protocol deviation, which does not meet the criteria for notification.
19) According to the protocol, a brain CT scan should be performed in the selection visit in order to exclude brain metastasis (exclusion criteria). The site used a previous version of the protocol where the CT scan wasn't required so 6 patients out of 10 were included without brain CT.	Yes , if this had an impact on patient safety.
20) A potential Serious Breach was identified, but not reported (documentation in the Sponsor's TMF identified that there may have been fraud at an investigator site, re-use of previous time point data in later time points). The Sponsor had investigated and the issue was subsequently found to be a genuine error and not fraud.	No , on this occasion. However, had this been identified as fraud impacting on the integrity of the data, then this should have been reported as Serious Breach.
21) Patient Information Leaflet and Informed Consent updated, but at one trial site this was not relayed to the patients until approximately 2-3 months after approval.	No , if this was not a systematic or persistent problem and if no harm to trial subjects resulted from the delay. Yes , if there was a significant impact on the integrity of trial subjects (e.g. there was key safety information not relayed to subjects in a timely manner).
22) The early destruction of investigator site files (i.e. the study had only been completed a year earlier)	Yes
23) The investigator would not allow any party access to the patients notes.	Yes , the data therefore could not be verified. The protocol would usually contain a clause to state that Sponsor representative and Regulatory authorities will have access to the data, and this is also reflected in the informed consent.
24) Loss of data due for example to a computer server's breakdown.	Yes , clinical trial sponsors and vendors should have agreements in place addressing business continuity and ensuring that clinical trials data are retrievable at any point in time.
25) Patients incorrectly randomized/stratified according to the protocol.	Yes , as this will be likely to have a significant impact on the data.
26) The Data and Safety Monitoring Board (DSMB)/ Data Monitoring Committees (DMC), which should be implemented according to the protocol and the clinical trial authorisation in a blinded trial, has in fact not been implemented.	Yes , the missing implementation of the DSMB/DMC has significant potential to impact the safety of trial subjects.
27) Patient's visit was not within the window period. Visit was supposed to be on day 28 (+/- 2 days), but subject had returned on day 24.	No , if there was no impact on the safety or integrity of trial subjects or on the scientific value of the trial.
28) Sampling for Pharmacokinetics (PK) parameter was collected five minutes out of permitted window period.	No , if there was no impact on the safety or integrity of trial subjects or on the scientific value of the trial. Yes , if the issue was systematic and persistent leading to a significant impact on the safety or integrity of subjects or scientific value of the trial.

29) Subjects did not complete the end-of-study questionnaires.	No , if there was no impact on the safety or integrity of trial subjects or on the scientific value of the trial.
30) The Interactive Voice Response System (IVRS) was used to randomize subjects and assign treatment kits to subjects. Some treatment kits shipped to the sites had misallocated treatment type in IVRS. As a result, subjects may have received misallocated treatment for some cycles.	Yes , there was significant potential to impact the safety or integrity of trial subjects and scientific value of the trial.
31) The label on the IMP did not reflect the correct concentration of IMP in the vial, resulting in subject receiving more than the required amount of IMP.	Yes , there was significant potential to impact the safety or integrity of trial subjects.
32) The study site failed to reduce or stop IMP dosing, in response to certain laboratory parameters or criteria, as required by the protocol. As a result, subject was exposed to an increased risk of adverse events.	Yes , there was significant potential to impact the safety or integrity of trial subjects.
33) The subject was to take the IMP daily for 30 days but instead took the IMP for 21 days; subject informed that he had forgotten to take the IMP for a week.	No , if there was no impact on the safety or integrity of trial subjects or on the scientific value of the trial. In addition, the assessment of the breach identified this as a single episode and a detailed corrective and preventive action plan was implemented.
34) Informed Consent was not obtained from a subject.	Yes , there was significant potential to impact the safety or integrity of the trial subject.
35) Subject was enrolled into the clinical trial, despite subject's (or his/ her legal representative's, as applicable) refusal (or objection) to participate in the trial.	Yes , there was significant potential to impact the safety or integrity of the trial subject.
36) Subject had consented on an Informed Consent Form (ICF) or substantial amendment to an ICF, which had never been reviewed nor approved by IRB/IEC.	Yes , there was significant potential to impact the safety or integrity of trial subjects.
37) The study investigator failed to ensure that the subject was eligible for the trial, prior to subject dosing. This resulted in an ineligible subject being exposed to the IMP and risk of adverse event.	Yes , there was significant potential to impact the safety or integrity of the trial subject.